



SAI NAGA KUMAR DEGALA

Signaling Engineer with over 7 years of Expertise in Indian Railway Signaling, specializing in Verification, Installation and Commissioning of TCAS (KAVACH) Systems. Proven ability in Signaling Systems Design, Testing and V&V of Electronic Subsystems, with hands-on Experience across more than 30 Station Installations. Currently expanding proficiency in UK Railway Signaling Principles and Modern Technologies, including TPWS and Axle Counter Systems. Adapt at Integrating complex Signaling Solutions, ensuring compliance with Technical Standards, and Delivering High-Quality, Safety-Critical Systems. Seeking to leverage deep Domain Knowledge and Adaptability to Contribute to advanced Railway Signaling Projects globally.

PERSONAL

- Name**
SAI NAGA KUMAR DEGALA
- Address**
Krishna District
521301 Gudivada
- Phone number**
09573864361
- Email**
saimahesh2667@gmail.com
- Date of birth**
05-07-1996
- Place of birth**
Visakhapatnam
- Gender**
Male
- Nationality**
India
- Marital status**
single
- LinkedIn**
www.linkedin.com/in/sai-naga-kumar-d-03b634198

INTERESTS

- Reading Technical Journals & Industry Updates
- Attending Cultural and Devotional Events
- Traveling & Exploring
- Sports & Fitness

LANGUAGES

- | | |
|---------|-------|
| Telugu | ★★★★★ |
| English | ★★★★★ |
| Hindi | ★★★★★ |



WORK EXPERIENCE

May 2025 - Present

Testing & Commissioning Engineer (CBTC) Alstom, Mumbai

CBTC / IXL Testing & Commissioning

- Performed **Pre-Acceptance Tests (PAT)** and **Site Acceptance Tests (SAT)** for CBTC Systems.
- Performed **CLC (IXL) Signaling System Testing** and verified **Vital/non-Vital I/O allocation charts**
- Produced **Relay Room layouts** and **Earthing arrangement plans**.
- Verify **Interface circuits** for Signals, Points, and Axle Counters.
- Performed **PAT Testing for KLCR, ESP Signal, and Points**

ATS Subsystem

- Ensured compliance with Signaling Standards during Commissioning.
- Executed **OSP and ATS baseline Software deployment and Configuration** on central servers using **VMware and ICONIS platforms**
- Performed **data migration and validation** for ZC, LC, CLC, and ATS during Testing phases.
- Monitored and managed **Level-1, Level-2, Gateway, FEP, and VDU servers** during Commissioning Activities.
- Configured **Cisco switches** and allocated station-level ports to support Signaling Networks.
- Basic knowledge of **ATC principles** and Integration with ATS Subsystems.

Jun 2022 - May 2025

Signalling Engineer REZNET RAIL PVT LTD, Hyderabad

Signalling Engineer – UK Design (Under Mentorship)

- Designed and replaced **UK Location Cases** (new and existing) in compliance with Signaling Standards.
- Enabled **TPWS circuits** and implemented **Earth strengthening** for Location Cases.
- Produced **Frauscher Axle Counter Interface Circuits, AEB card data, and COM card data**.
- Developed **TSR & ESR Designs** and updated final records for **ATP recovery**.
- Created **HABD designs** and performed **Power Calculations** for location Equipment.
- Prepared **Existing Recovery Designs, Spot wire Designs, and closer list copies**.
- Produced **Location Case Designs** (new & existing), **Control Tables, Commissioning copies, and AFC & AIR documentation**.
- Knowledge of **TPWS, AWS, HABD, and ATP systems**
- Designed **Smart Lock solutions** and replacement plans for **Axle Counter cables** (Location-to-Location and REB).

Indian Railways – TCAS (KAVACH)

- Designed and Developed **TCAS (KAVACH) layouts, SST control tables,**

- and **RFID tag data** for deployments.
- Executed **RFID tag Programming and uploading** including preparation of **RFID layouts and tag data**
- Performed **Gradient Calculations** and prepared **OHE (Overhead Equipment) data** for system integration.
- Performed **TOC testing**, correspondence tests, and commissioning activities.
- **KAVACH Systems Design**

○ Apr 2019 - May 2022

Service Engineer

MEDHA SERVO DRIVES PVT LTD, HYDERABAD

Responsibilities & Achievements

- Prepared **Cable Route Plans** and modified Logic alterations as per Customer requirements.
- Led **System Integration Activities**, including successful completion of **Factory Acceptance Tests (FAT)** and **Site Acceptance Tests (SAT)**.
- Correlated Circuits according to Site conditions and verified **Vital/non-Vital input-output Allocation Charts**.
- Performed **relay contact analysis**, prepared **Cable Termination Plans (CT/Relay Rack)**, and produced **Signaling Relay room layouts** with **Earthing arrangement Plans**.
- Verify **Interface Circuits** for Signals, Points, Axle Counters, DC track Circuits, and Level Crossing Gate Circuits.
- Performed **Load Calculations**, verified **Bit Charts**, and Tested **Control Tables and Square sheets**.
- Performed **Bel tests** for Wiring Verification and Ensured Compliance with **Technical Specifications, Signaling Principles, RDSO guidelines, and latest Technical Advisory Notes (TAN)**.
- Executed **Installation, Testing, and Commissioning of TCAS (KAVACH) systems**, including **RFID Tag Programming and uploading**
- Verify **EI Interface Wiring** performed **Bell testing**, and analyzed TOC and correspondence tests jointly with railway staff.
- Performed **Site surveys for TCAS**, verified system power supply, performed power-on and load testing.
- Improved installation standards and prepared **floor plans based on site conditions**.
- Worked with communication protocols such as **RS232, RS485, and Fiber Optic Communication (OFC)**.
- Successfully completed **Installation, Testing, and Commissioning of TCAS (KAVACH) systems across 30 stations**.



SKILLS

Testing and commissioning	★ ★ ★ ★ ★
design engineer	★ ★ ★ ★ ★
signalling engineer	★ ★ ★ ★ ★
Verification and Validation	★ ★ ★ ★ ★
MicroStation	★ ★ ★ ★ ★
Troubleshooting	★ ★ ★ ★ ★
Railway signalling systems	★ ★ ★ ★ ★



EDUCATION AND QUALIFICATIONS

○ Jun 2011 - Apr 2014

VKR & VNB POLYTECHNIC
Gudiavda



PUBLICATIONS

Professional Summary

- Enthusiastic and quick learner with a strong desire to embrace challenges and acquire new concepts rapidly.
- Resourceful team player skilled in building trust and maintaining positive relationships with colleagues and clients.
- Highly motivated self-starter who takes initiative and works effectively with minimal supervision.
- Dedicated and productive professional with a solid work ethic, consistently delivering quality results.
- Strong time management skills, ensuring tasks are completed efficiently and within deadlines.

Declaration

I hereby declare that the above information is true to the best of my knowledge. I pledge to work sincerely and diligently, adhering to organizational standards and contributing to its success.